In [4]:
```
import pandas as pd

data = pd.read_csv('bankmarket.csv', sep=';')
data
```

Out[4]:

	age	job	marital	education	default	housing	loan	cont
0	56	housemaid	married	basic.4y	no	no	no	teleph
1	57	services	married	high.school	unknown	no	no	teleph
2	37	services	married	high.school	no	yes	no	teleph
3	40	admin.	married	basic.6y	no	no	no	teleph
4	56	services	married	high.school	no	no	yes	teleph
...	...	...	...	...	...	...	...	
41183	73	retired	married	professional.course	no	yes	no	cell
41184	46	blue-collar	married	professional.course	no	no	no	cell
41185	56	retired	married	university.degree	no	yes	no	cell
41186	44	technician	married	professional.course	no	no	no	cell
41187	74	retired	married	professional.course	no	yes	no	cell

41188 rows × 21 columns

In [5]:
```
features = ['age', 'duration', 'campaign', 'previous', 'housing', 'loan']
X = data[features].copy()
y = data['y']

X['housing'] = X['housing'].map({'yes': 1, 'no': 0, 'unknown': -1})
X['loan']    = X['loan'].map({'yes': 1, 'no': 0, 'unknown': -1})
X
```

Out[5]:

	age	duration	campaign	previous	housing	loan
0	56	261	1	0	0	0
1	57	149	1	0	0	0
2	37	226	1	0	1	0
3	40	151	1	0	0	0
4	56	307	1	0	0	1
...	...	...	...	...	...	...
41183	73	334	1	0	1	0
41184	46	383	1	0	0	0
41185	56	189	2	0	1	0
41186	44	442	1	0	0	0
41187	74	239	3	1	1	0

41188 rows × 6 columns

In [6]:
```
from sklearn.model_selection import train_test_split

X_train, X_temp, y_train, y_temp = train_test_split(X, y, test_size=0.30,
X_val, X_test, y_val, y_test     = train_test_split(X_temp, y_temp, test_

print(f'Train : {X_train.shape[0]}')
print(f'Val   : {X_val.shape[0]}')
print(f'Test  : {X_test.shape[0]}')
```

```
Train : 28831
Val   : 6178
Test  : 6179
```

In [7]:
```
from sklearn.ensemble import RandomForestClassifier

rf = RandomForestClassifier(random_state=42)
rf.fit(X_train, y_train)

rf_val_acc  = rf.score(X_val, y_val)
rf_test_acc = rf.score(X_test, y_test)

print(f'Random Forest - Validation Accuracy : {rf_val_acc:.4f}')
print(f'Random Forest - Test Accuracy       : {rf_test_acc:.4f}')
```

```
Random Forest - Validation Accuracy : 0.8851
Random Forest - Test Accuracy       : 0.8854
```

In [8]:
```
from sklearn.ensemble import AdaBoostClassifier

ab = AdaBoostClassifier(random_state=42)
ab.fit(X_train, y_train)

ab_val_acc  = ab.score(X_val, y_val)
ab_test_acc = ab.score(X_test, y_test)
```

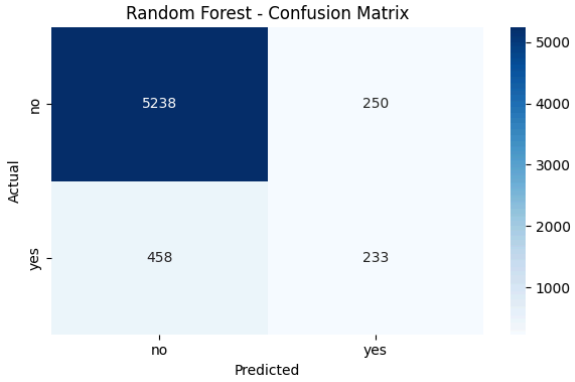
```
print(f'AdaBoost - Validation Accuracy : {ab_val_acc:.4f}')
print(f'AdaBoost - Test Accuracy       : {ab_test_acc:.4f}')
```

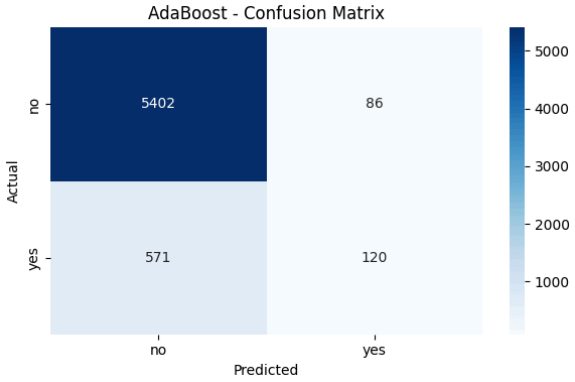
```
AdaBoost - Validation Accuracy : 0.8935
AdaBoost - Test Accuracy       : 0.8937
```

In [9]:
```
from sklearn.metrics import classification_report, confusion_matrix

print('=== Random Forest ===')
print(classification_report(y_test, rf.predict(X_test)))

print('=== AdaBoost ===')
print(classification_report(y_test, ab.predict(X_test)))
```

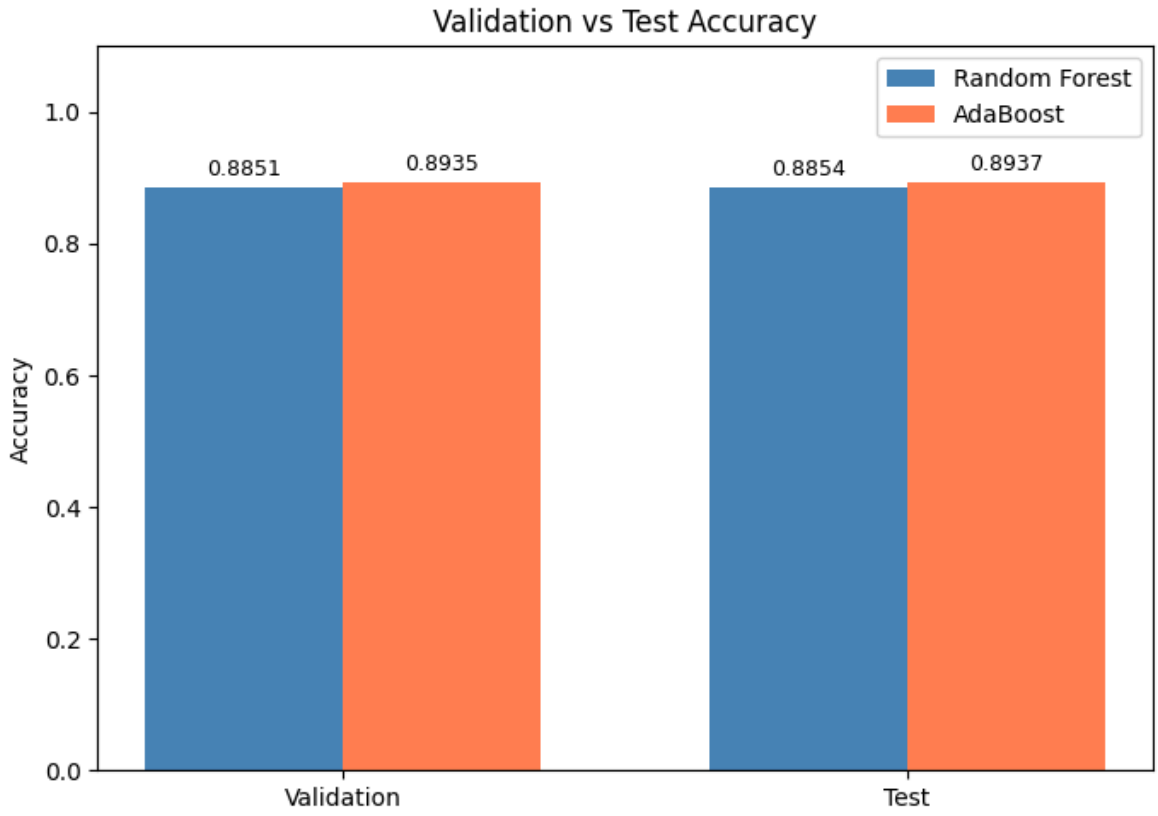
```
=== Random Forest ===
              precision    recall  f1-score   support

          no       0.92      0.95      0.94      5488
         yes       0.48      0.34      0.40       691

    accuracy                           0.89      6179
   macro avg       0.70      0.65      0.67      6179
weighted avg       0.87      0.89      0.88      6179

=== AdaBoost ===
              precision    recall  f1-score   support

          no       0.90      0.98      0.94      5488
         yes       0.58      0.17      0.27       691

    accuracy                           0.89      6179
   macro avg       0.74      0.58      0.61      6179
weighted avg       0.87      0.89      0.87      6179
```

In [10]:
```
import matplotlib.pyplot as plt
import seaborn as sns

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

for ax, model, name in zip(axes,
                           [rf, ab],
                           ['Random Forest', 'AdaBoost']):
    cm = confusion_matrix(y_test, model.predict(X_test))
    sns.heatmap(cm, annot=True, fmt='d', ax=ax, cmap='Blues',
                xticklabels=['no','yes'], yticklabels=['no','yes'])
    ax.set_title(f'{name} - Confusion Matrix')
    ax.set_xlabel('Predicted')
    ax.set_ylabel('Actual')

plt.tight_layout()
plt.show()
```





In [11]:
```
labels  = ['Validation', 'Test']
rf_scores = [rf_val_acc, rf_test_acc]
ab_scores = [ab_val_acc, ab_test_acc]

x = range(len(labels))
width = 0.35

fig, ax = plt.subplots(figsize=(7, 5))
b1 = ax.bar([i - width/2 for i in x], rf_scores, width, label='Random For
b2 = ax.bar([i + width/2 for i in x], ab_scores, width, label='AdaBoost',

ax.set_ylim(0, 1.1)
ax.set_xticks(x)
ax.set_xticklabels(labels)
ax.set_ylabel('Accuracy')
ax.set_title('Validation vs Test Accuracy')
ax.legend()

for bar in list(b1) + list(b2):
    ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.01,
            f'{bar.get_height():.4f}', ha='center', va='bottom', fontsize

plt.tight_layout()
plt.show()
```


Validation vs Test Accuracy
Random Forest
AdaBoost
0.8851
0.8935
0.8854
0.8937
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Validation
Test